Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/28/2007
PI Name:	Good, Edward F M.D.		
Project Title:	The Role of Intracranial Pressure in Space Adaptation Syndrome		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77598	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	01/01/2001	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	Shuttle		
Flight Assignment:	TBD In flight development phase (not yet manifested) Original end date was 3/15/2009.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ04HD94A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space Adaptation Syndrome (SAS) affects space flight crews through loss of productivity, physical discomfort and safety issues. The headward fluid shift associated with the launch posture and early exposure to weightlessness would be expected to increase intracranial pressure (ICP), which in turn can produce symptoms associated with SAS, particularly headache and vomiting. We propose to measure ICP non-invasively in crewmembers before, during and after brief Space Shuttle missions. We will use two complementary techniques to assure that we identify and characterize any changes in ICP accurately: the Marchbanks Cerebral and Cochlear Fluid Pressure (CCFP) Analyzer and the Pulsed Phase-Lock Loop (PPLL) Ultrasound technique developed by W. T. Yost and J. H. Cantrell at NASA Langley Research Center and A. R. Hargens' group at NASA Ames Research Center. These ICP measurements will be compared with the time course and extent of SAS symptoms, quantified according to the Graybiel motion-sickness scale. If changes in ICP are found to be associated with SAS, then countermeasures that shift ICP toward ground-based norms could be effective in eliminating or reducing the symptoms of SAS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A better understanding of time-course changes in ICP during exposure to microgravity could aid our understanding of the pathophysiology of Space Adaptation Syndrome so that treatment and performance of astronauts could be improved during early flight. Considerable Earth benefit also would be derived: The potential improvement in understanding the relationships between increased ICP and headache, motion sickness and cognitive performance without the confounding effects of injury. Early noninvasive measurements of ICP may help reduce the mortality and morbidity associated with cerebral disease.
Task Progress:	No progress report received this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)